Probability
Lecture 7: Linearity of Expectation

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Content

1. Linearity of Expectation
2. Portfolio Selection

Linearity of Expectation

$$\mathbb{E}[X + Y] = E[X] + \mathbb{E}[Y].$$

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + 2\mathrm{Cov}(X, Y) + \mathrm{Var}(Y).$$

Theorem

If $X = X_1 + X_2 + \ldots + X_n$, *then*

$$\mathbb{E}[X] = \mathbb{E}[X_1 + X_2 + \ldots + X_n] = \mathbb{E}[X_1] + \mathbb{E}[X_2] + \ldots + \mathbb{E}[X_n].$$

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First Example

- Let $p_n(k)$ be the number of permutations from 1 to n having exactly k fixed points. Find the value of

$$\sum_{k=1}^{n} k p_n(k).$$

+ Draw a random permutation by uniform distribution.
+ Denote X by the number of fixed points of the permutation.

X	0	1	...	n
$\mathbb{P}$	$p_n(0)/n!$	$p_n(1)/n!$	...	$p_n(n)/n!$

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Example

+ By the definition of expectation,

$$\mathbb{E}[X] = \sum_{k=0}^{n} kp_n(k)/n! = \sum_{k=1}^{n} kp_n(k)/n!.$$

+ On other hands, express $X = I_1 + I_2 + \ldots + I_n$, where I_j is the indicator function that whether i is a fixed point or not.

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{P}(i \text{ fixed}) = \sum_{k=1}^{n} \frac{(n-1)!}{n!} = 1.$$

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Second Example

- Toss a fair coin until TH occurs. What is the expected number of tossing?

Idea: we toss until the first T appears, and from then, we wait for the first appearance of H.

$$X = Geo(T) + Geo(H).$$

$$\mathbb{E}X = EGeo(T) + \mathbb{E}Geo(H) = 2 + 2 = 4.$$

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Third Example (Quant job interview)

- Toss a fair coin until HH occurs. What is the expected number of tossing? - Idea:

- If the first toss is T, then nothing special happens, and we have a new beginning.
- If the first toss is H, then depending on the second toss, we have
  - If the second toss is H again, then we stop here after 2 tosses.
  - If the second toss is T, back to nothing special happens, and we have a new beginning.

$$E = \frac{1}{2}(1+E) + \frac{1}{2}\left[\frac{1}{2}2 + \frac{1}{2}(2+E)\right]$$

Hence $E = 6$.

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Fourth Example

- Liams bowl of spaghetti contains n strands. He selects two ends at random and joins them together. He does this until there are no ends left. What is the expected number of spaghetti loops in the bowl?

- Idea:

- If in the first join, we make a loop, then the number of loops is 1+ the one in $n-1$ strands case.
- If in the first join, we do not make a loop, then we back to the $n-1$ strands case.

$$E_n = \frac{n}{C_{2n}^2}(1+E_{n-1}) + \left(1 - \frac{n}{C_{2n}^2}\right) E_{n-1} = \frac{1}{2n-1} + E_{n-1}.$$

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Fifth Example

- Ten boys and ten girls stand in row. What is the expectation of the number of couples of a boy and a girl standing next to each other?

- Idea:

$$X = I_1 + \ldots + I_{19},$$

where I_k is an indicator function that at the i-th gap, whether there exists a couple of a boy and a girl.

$$\mathbb{P}(I_i = 1) = \frac{2.5.5.18!}{20!} = \frac{5}{38},$$

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Buffon needlet problem

- Suppose we have a floor made of parallel strips of wood of unit width. We randomly drop a needlet of length l onto the floor. What is the probability that the needle will lie across a line between two strips?

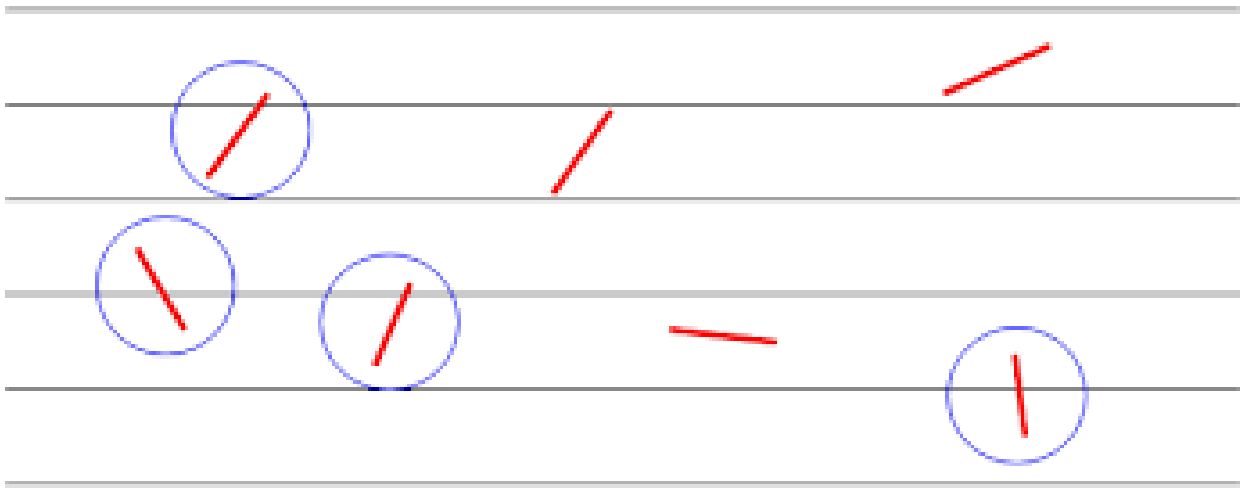


Figure: Numerical simulation of Buffon problem

Buffon needlet problem

- Let X_l be the number of intersections between a needlet of length l with the strips.
- Let's break into two smaller needlets of lengths $l1$ and $l2$.
- Since $X_l = X_{l1} + X_{l2}$ almost surely,

$$\mathbb{E}[X_l] = \mathbb{E}[X_{l1}] + \mathbb{E}[X_{l2}];$$

in general, if $l_1 + \ldots + l_k = l$ then

$$\mathbb{E}[X_l] = \mathbb{E}[X_{l_1}] + \mathbb{E}[X_{l_2}] + \ldots + \mathbb{E}[X_{l_k}].$$

- It deduces that there exists a constant c such that

$$\mathbb{E}[X_l] = c.l$$

- How to find c?
$\Rightarrow$ Consider a circle with radius $1/2$ or perimeter π. It always has two intersection points with the strips. $2 = c\pi \Rightarrow c = 2/\pi$.

1 Linearity of Expectation

2 Portfolio Selection

Markowitz theory

- Proposed by Markowits since 1950.
- Nobel prize 1990 in Economics, nominated to Markowitz, Sharpe and Miller.
- Model: consider in market, there exist n kinds of risky assets (or stocks). Basic information on the return rate R_i's

$$\mathbb{E}[R_i] = r_i,\ DR_i = \sigma_i^2,\ \rho_{ij} = corr(R_i, R_j) = \frac{Cov(R_i, R_j)}{\sqrt{D(R_i)D(R_j)}}.$$

- A portfolio is a way to divide the capital into the stocks, with weights w_i

$$w_1 + w_2 + \ldots + w_n = 1.$$

- If you allow $w_i < 0$, we will call **sell short**.

Sell short

- In practice, use when you predict that the price will go down.
- Example:
  + Assume that you have 1\$ and the price of the stock is 39\$.
  + Sell short: you **borrow** one stock from the broker and sell it to obtain 39\$.
  + Now you have $1 + 39 = 40$\$. This amount is an assurance for the stock you borrowed. It means that if the price of stock is less than 40\$ then nothing will happen, but else if it goes to 40\$, then the broker immediately intercept your account to use 40\$ to buy a stock for your loan.
- Related operation: margin.

Risk

- **Risk: Variance**.

Theorem

The return rate of a portfolio is

$$R_V = w_1 R_1 + \ldots + w_n R_n.$$

Then

$$\mathbb{E}R_V = w_1 r_1 + \ldots + w_n r_n,$$

$$DR_V = \sigma_V^2 = \sum_{i,j=1}^{n} w_i w_j . \sigma_i \sigma_j \rho_{ij}.$$

- Two questions:
  - Find the portfolio with smallest risk?
  - Among the portfolios of a given return rate, find the one with smallest risk?

Portfolio Selection

By Lagrange multiplier method

$$L(w_1, \ldots, w_n, \lambda) = \sum_{i,j=1}^{n} w_i w_j . \sigma_i \sigma_j \rho_{ij} + \lambda(w_1 + w_2 + \ldots + w_n - 1).$$

Theorem

The portfolio with smallest risk is

$$w = \frac{u\Sigma^{-1}}{u\Sigma^{-1}u^T},$$

with $u = (1, \ldots, 1)$ *and* Σ*: covariance matrix of the random vector* $(R_1 \ldots, R_n)$.

Portfolio Selection

$$L(w_1, \ldots, w_n, \lambda, \theta) = \sum_{i,j=1}^{n} w_i w_j . \sigma_i \sigma_j \rho_{ij} + \lambda(w_1 + w_2 + \ldots + w_n - 1) + \theta(w_1 r_1 + \ldots + w_n r_n - \mu).$$

Theorem

Fixed the expectation return rate μ, the portfolio with smallest risk is

$$w = \frac{\begin{vmatrix} 1 & u\Sigma^{-1}m^T \\ \mu & m\Sigma^{-1}m^T \end{vmatrix} u\Sigma^{-1} + \begin{vmatrix} u\Sigma^{-1}u^T & 1 \\ m\Sigma^{-1}u^T & \mu \end{vmatrix} m\Sigma^{-1}}{\begin{vmatrix} u\Sigma^{-1}u^T & u\Sigma^{-1}m^T \\ m\Sigma^{-1}u^T & m\Sigma^{-1}m^T \end{vmatrix}},$$

with $m = (r_1, \ldots, r_n)$.

Efficient frontier

Definition

Consider two portfolios w and w'. We say that w is more efficient than w' if

$$\mu_w \geq \mu_{w'}; \quad \sigma_w \leq \sigma_{w'},$$

Definition

Consider the coordinate plane (σ, μ) of the portfolios, we can define the **efficient frontier** that is the set of all possible portfolios such that there does not exist any portfolios that is more efficient than them.

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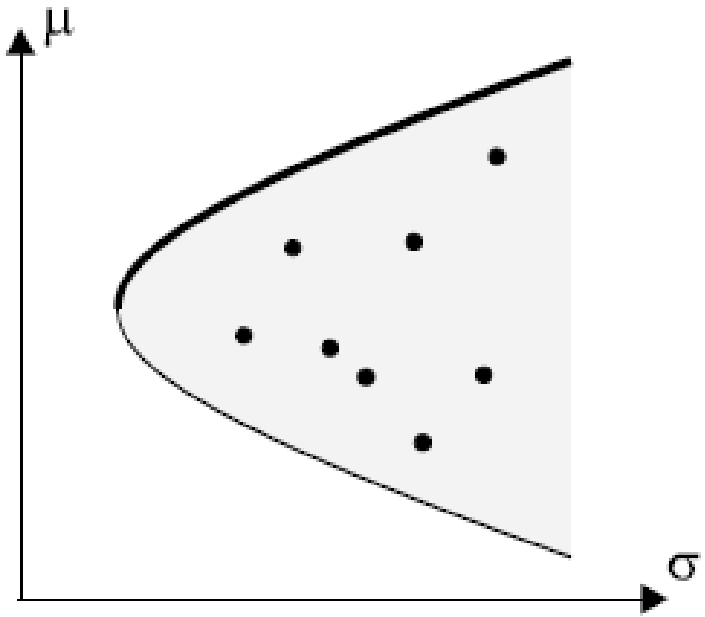
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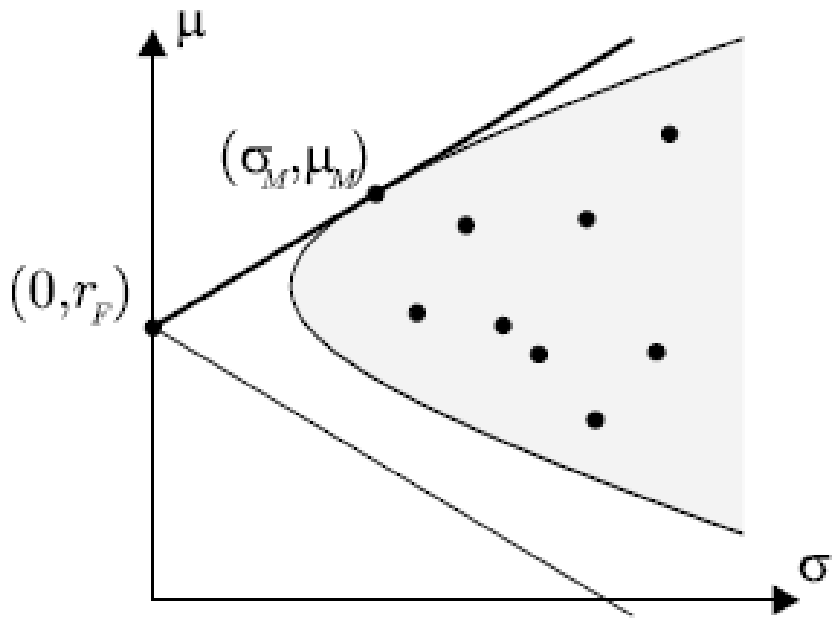
Theorem

The efficient frontier is convex. It means that for three points $(\sigma_1, \mu_1), (\sigma_2, \mu_2)$ *and* (σ_3, μ_3) *on the frontier such that* $\mu_2 = p\mu_1 + (1-p)\mu_3$ *with* $p \in [0,1]$ *then*

$$\sigma_2 \leq p\sigma_1 + (1-p)\sigma_3.$$

Markowitz point

- Let us now consider one more kind of asset: **risk-free asset**, for instance: bank savings. That is the point $(0, r_f)$.
- Draw a tangent line from this point to the efficient frontier.



Theorem

This line is the new **efficient frontier** *for portfolios with* $n+1$ *stocks.*

Thank you!